Data Pipeline for Large-Scale TV Advertising and Content Analysis

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ROLE AND TASKS

Processing and organizing large-scale television viewership and advertising data to enable analysis of media consumption patterns

• **Data Infrastructure Development** – Built TV network reference database with 400+ networks and implemented fuzzy matching system to identify network self-promotional ads with alternate name handling

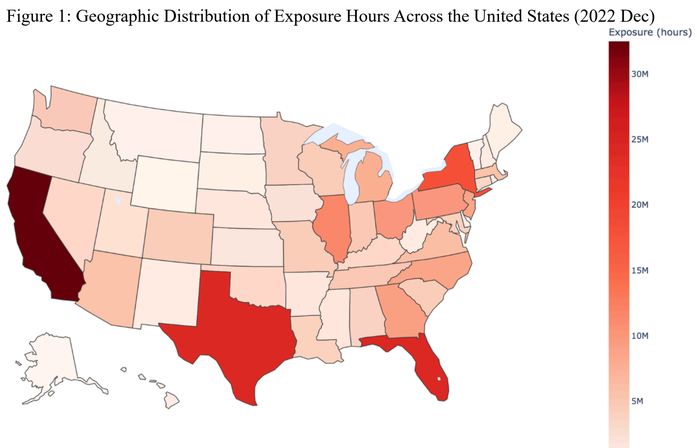
• **Large-Scale Data Processing** – Processed over 29 billion device-ad viewing records and cleaned viewership data by removing unnecessary columns and merging fragmented sessions into complete records with corrected timestamps

GENERAL STATISTICS

Table 1: Quarterly panel summary (example). Replace caption as needed.

Quarter	Days with viewing data	Total viewing (million h)	Unique Users (thousands)	Unique ZIP codes
2022 Q4	92	533.14	7,647.7	75,180
2023 Q1	90	589.97	7,456.0	75,415
2023 Q2	91	527.95	7,330.9	76,169
2023 Q3	92	522.48	7,159.4	76,907
2023 Q4	92	514.60	7,068.3	76,852
2024 Q1	91	612.64	7,433.0	76,227
2024 Q2	91	483.81	6,976.9	78,360

Figure 1: Geographic Distribution of Exposure Hours Across the United States (2022 Dec)



INPUT AND OUTPUT

Input:

- **TV Ad Exposure Files** – gzip-compressed CSV files containing device-ad watch events
- **TV Content Viewership Files** – gzip-compressed CSV files containing device-content viewing records across multiple dates (August 2022 – July 2024)
- **Data Dictionary PDF** – Reference documentation describing table structure and variable definitions

Output:

- **TV Network Reference List** – CSV file created with 400+ networks, including names, types (TV/Cable/Streaming), genres, alternate names, and parent companies
- **Filtered Ad Exposure Output** – CSV files containing only rows where advertisers are TV networks
- **Collapsed Viewership Output** – CSV files with aggregated viewing sessions

SKILLS AND TOOLS

Programming & Software:

- **Python:** Pandas (data manipulation), NumPy (numerical operations), regex (pattern matching), gzip (file decompression), shutil (file operations), glob (file finding), os (directory navigation), datetime (time tracking), tqdm (progress bars), pathlib (path handling), matplotlib (data visualization)
- **Software/Platform**: Google Colab*, Google Drive, IDEs(Pycharm, VSCode), OpenRefine, Cursor
- **Data Processing & Analysis**: Large-scale dataset management (100M+ rows), Data cleaning and transformation, Row aggregation and consolidation, Memory-efficient processing

Recommended Courses:

- **Economics**: **ECN-345-Prediction Markets**, ECN-372-Prediction and Machine Learning in Economics
- **Statistics**: **STA-261-Statistical Computing**, STA-363-Statistical Learning

CHALLENGES AND SOLUTIONS

Dataset exceeded local memory constraints and could not be processed on a local machine	• implemented in Google Colab with data accessed directly from Google Drive, reducing local storage and memory load
Silent Data Loss due to Memory Constraint/Incomplete Processing	• splitting the output into eight smaller files instead of two, reducing memory and write pressure during processing

VISUALIZATIONS

Figure 2. Daily Exposure Hours on Selected Networks (Jan 2023 - Mar 2023)

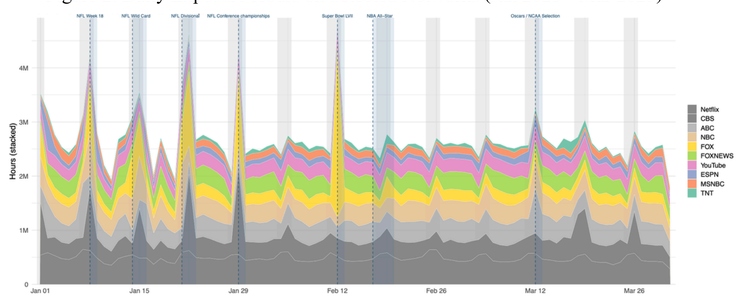
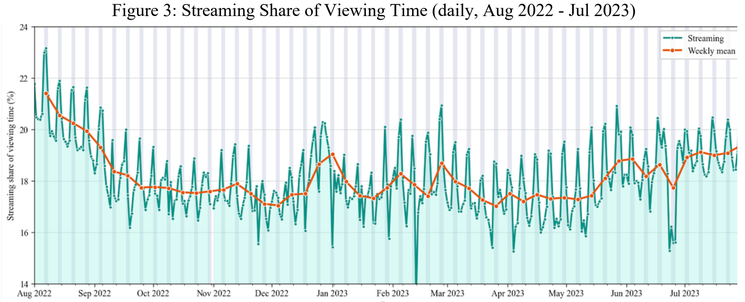



Figure 3: Streaming Share of Viewing Time (daily, Aug 2022 - Jul 2023)